EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Lab Examination, Summer 2026 Semester

Course:	CSE 207- Data Structures, Section-1, Set-A
Instructor:	Dr. Maheen Islam, Associate Professor, CSE Department
Full Marks:	20
Time:	1 Hour

1. Write a program to represent and manipulate two sets of integers using doubly linked lists. Each set should be stored in a separate doubly linked list. Since a set cannot contain duplicate elements, the program must ensure that each value appears only once in a set. Each node of the doubly linked list should contain an integer value, a pointer to the previous node, and a pointer to the next node.

Tasks

1. **Input Two Sets**
   - Write a function to input the elements of the first set and store them in a doubly linked list.
   - Write another function to input the elements of the second set.
   - Duplicate values should not be inserted into a set.
2. **Display the Sets**
   - Write a function to display the elements of each set by traversing the doubly linked list from beginning to end.
3. **Find the Union**
   - Write a function to find the **union** of the two sets.
   - The union should contain every distinct element that appears in either set.
   - Store the result in a new doubly linked list.
4. **Find the Symmetric Difference**
   - Write a function to find the **symmetric difference** of the two sets.
   - The result should contain all elements that are present in either Set A or Set B, but **not in both.**
   - Store the result in another doubly linked list.

Sample Input and Output

Input:

Enter the number of elements in Set A: 5
Enter the elements:10 20 30 40 50
Enter the number of elements in Set B: 4
Enter the elements:20 40 60 80

Output:

Set A:
10 20 30 40 50
Set B:
20 40 60 80
Union:
10 20 30 40 50 60 80
Symmetric Difference:
10 30 50 60 80

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EAST WEST UNIVERSITY
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Lab Examination, Summer 2026 Semester

Course:	**CSE 207- Data Structures, Section-1, Set-B**
Instructor:	**Dr. Maheen Islam, Associate Professor, CSE Department**
Full Marks:	**20**
Time:	**1 Hour**

Write a program to represent and manipulate **two sets of integers using doubly linked lists.** Each set should be stored in a separate doubly linked list. Since a set cannot contain duplicate elements, the program must ensure that each value appears only once in a set.

Each node of the doubly linked list should contain an integer value, a pointer to the previous node, and a pointer to the next node.

Tasks

1. **Input Two Sets**
   - Write a function to input the elements of the first set and store them in a doubly linked list.
   - Write another function to input the elements of the second set.
   - Duplicate values should not be inserted into a set.
2. **Display the Sets**
   - Write a function to display the elements of each set by traversing the doubly linked list from beginning to end.
3. **Find the Difference**
   - Write a function to find the **difference A − B.**
   - The result should contain all elements that are present in Set A but not in Set B.
   - Store the result in a new doubly linked list.
4. **Find the Intersection**
   - Write a function to find the **intersection** of the two sets.
   - The intersection should contain only those elements that are present in both sets.
   - Store the result in another doubly linked list.

Sample Input and Output

Input:

Enter the number of elements in Set A: 5
Enter the elements:
10 20 30 40 50
Enter the number of elements in Set B: 4
Enter the elements:
20 40 60 80

Output:

Set A:
10 20 30 40 50
Set B:
20 40 60 80
Difference (A - B):
10 30 50
Intersection:
20 40

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Time:	1 Hour

Write a program to represent and manipulate a polynomial using a singly linked list.
Each node of the linked list should contain the coefficient of a term, the exponent of the term, and a pointer to the next node.

Tasks

1. **Input the Polynomial**
   - Write a function to input the terms of a polynomial and store them in a singly linked list.
   - Each term should contain a coefficient and an exponent.
2. **Display the Polynomial**
   - Write a function to display the polynomial in standard polynomial form.
3. **Differentiate the Polynomial**
   - Write a function to find the first derivative of the polynomial.
   - For each term, the differentiated term should be.
   - Constant terms should not appear in the resultant polynomial.
   - Store the derivative in a new linked list.
4. **Display the Resultant Polynomial**
   - Write a function to display the differentiated polynomial.

Sample Input and Output

Input:

Enter the number of terms in the polynomial: 4
Enter coefficient and exponent:
5 4
3 3
2 1
7 0

Output:

Polynomial:
5x^4 + 3x^3 + 2x + 7

Derivative:
20x^3 + 9x^2 + 2